

BSU Cyber Squad Camp 2026

Devharsh Trivedi, Ph.D., CISSP | Department of Computer Science, Bowie State University

Five-day cybersecurity camp for Bowie State University students.

No prior experience needed.

What you will learn

- Day 1** Python basics and how the internet works
- Day 2** CIA triad, passwords, and account security
- Day 3** Cryptography: ciphers, hashing, and TLS
- Day 4** Web security: SQL injection, XSS, OWASP
- Day 5** CTF competitions, blue team, and careers

What you need

A laptop and a browser.
Everything runs in the cloud.

Course website:

bsu-devharsh.github.io/Cyber-Squad

Google Colab (no install needed)
U.S. Cyber Range credentials
provided at each lab session



How to use

- Open your day deck at bsu-devharsh.github.io/Cyber-Squad
- On the Python slide, click Colab -- no install needed
- For Cyber Range labs, get credentials from instructor
- Each day ends with pair work in the student notebook

Cyber Range environments

Days 1-2:

Cyber Basics (2025.7)

Days 3-5:

K12 Cyber Security Concepts
Using Kali Linux (2025.7)

Instructor shares direct links
at the start of each lab session.

1

Linux and Networking Foundations

Variables, loops, functions, Linux filesystem, OSI model, subnetting. US Cyber Range: Cyber Basics.

Daily Schedule

Time	Block	Slides
9:30 - 10:55	Lecture 1	Parts 1-2
10:55 - 11:05	Break (10 min)	
11:05 - 12:30	Lecture 2 + Python Exercise	Parts 3+6
12:30 - 1:30	Lunch	
1:30 - 2:25	Cyber Range lab	Part 4
2:25 - 3:20	Interactive game	Part 7
3:20 - 4:20	Pair work / Presentations	Part 9
4:20 - 4:30	Wrap up	Part 8

Lecture 1 (9:30-10:55)

Get oriented and learn the basics. Part 1: Get oriented (3 slides). Part 2: Linux, networking, CLI, SSH (14 slides).

Lecture 2 (11:05-12:30)

Python exercise intro, Cyber Range walkthrough, knowledge checks. Parts 3, 4, and 6.

Cyber Range (1:30-2:25)

Cyber Basics (2025.7) -- VM login, CLI navigation, SSH into remote host.

Interactive Game (2:25-3:20)

Part 5 -- Network packet simulation game. Students build and inspect packets.

Pair Work (3:20-4:20)

Day 1 coding challenge in Day1_student.ipynb. Work with your partner on CLI scripts.

Wrap Up (4:20-4:30)

Recap Linux and networking. Preview Day 2: passwords and hashing.

2

Accounts, Passwords, and Hashing

CIA triad, threat actors, authentication, password entropy, hashing with salt. US Cyber Range: Cyber Basics.

Daily Schedule

Time	Block	Slides
9:30 - 10:55	Lecture 1	Parts 1-2
10:55 - 11:05	Break (10 min)	
11:05 - 12:30	Lecture 2 + Python Exercise	Parts 3+6
12:30 - 1:30	Lunch	
1:30 - 2:25	Cyber Range lab	Part 4
2:25 - 3:20	Interactive game	Part 7
3:20 - 4:20	Pair work / Presentations	Part 9
4:20 - 4:30	Wrap up	Part 8

Lecture 1 (9:30-10:55)

Get oriented and security foundations. Part 2: CIA triad, threat actors, authentication (15 slides).

Lecture 2 (11:05-12:30)

Password entropy math, keyspace calculation, Python exercise intro. Parts 3, 4, and 6.

Cyber Range (1:30-2:25)

Cyber Basics (2025.7) -- Password cracking, hash comparison, salt demo.

Interactive Game (2:25-3:20)

Part 5 -- Password strength game and threat scenario quiz.

Pair Work (3:20-4:20)

Build a keyspace calculator and hash cracker in Day2_student.ipynb.

Wrap Up (4:20-4:30)

Recap CIA triad and hashing. Preview Day 3: cryptography.

Daily Schedule

Time	Block	Slides
9:30 - 10:55	Lecture 1	Parts 1-2
10:55 - 11:05	Break (10 min)	
11:05 - 12:30	Lecture 2 + Python Exercise	Parts 3+6
12:30 - 1:30	Lunch	
1:30 - 2:25	Cyber Range lab	Part 4
2:25 - 3:20	Interactive game	Part 7
3:20 - 4:20	Pair work / Presentations	Part 9
4:20 - 4:30	Wrap up	Part 8

Lecture 1 (9:30-10:55)

History of cryptography and symmetric ciphers.
Part 2: Caesar, Base64, XOR (16 slides).

Lecture 2 (11:05-12:30)

RSA walkthrough, TLS handshake, frequency analysis, Python exercise. Parts 3, 4, and 6.

Cyber Range (1:30-2:25)

Kali Linux (2025.7) -- Capture and decrypt traffic, frequency analysis challenge.

Interactive Game (2:25-3:20)

Part 5 -- Cipher challenge game and frequency analysis puzzle.

Pair Work (3:20-4:20)

Implement Caesar cipher, XOR encoder, and Base64 decoder in Day3_student.ipynb.

Wrap Up (4:20-4:30)

Recap cipher types and TLS. Preview Day 4: web security.

Daily Schedule

Time	Block	Slides
9:30 - 10:55	Lecture 1	Parts 1-2
10:55 - 11:05	Break (10 min)	
11:05 - 12:30	Lecture 2 + Python Exercise	Parts 3+6
12:30 - 1:30	Lunch	
1:30 - 2:25	Cyber Range lab	Part 4
2:25 - 3:20	Interactive game	Part 7
3:20 - 4:20	Pair work / Presentations	Part 9
4:20 - 4:30	Wrap up	Part 8

Lecture 1 (9:30-10:55)

Web app architecture, URLs, client-server model. Part 2: SQL injection and XSS (16 slides).

Lecture 2 (11:05-12:30)

OWASP Top 10, session hijacking, AI prompt injection, Python exercise. Parts 3, 4, and 6.

Cyber Range (1:30-2:25)

Kali Linux (2025.7) -- SQLi and XSS on a vulnerable web app, Burp Suite intro.

Interactive Game (2:25-3:20)

Part 5 -- OWASP attack and defend simulation game.

Pair Work (3:20-4:20)

URL parser, input validator, and prompt injection guard in Day4_student.ipynb.

Wrap Up (4:20-4:30)

Recap OWASP Top 10 and AI attacks. Preview Day 5: CTF.

Daily Schedule

Time	Block	Slides
9:30 - 10:55	Lecture 1	Parts 1-2
10:55 - 11:05	Break (10 min)	
11:05 - 12:30	Lecture 2 + Python Exercise	Parts 3+6
12:30 - 1:30	Lunch	
1:30 - 2:25	Cyber Range lab	Part 4
2:25 - 3:20	Interactive game	Part 7
3:20 - 4:20	Pair work / Presentations	Part 9
4:20 - 4:30	Wrap up	Part 8

Lecture 1 (9:30-10:55)

CTF categories, OSINT methodology, blue team roles. Part 2: incidents, logs, OSINT (13 slides).

Lecture 2 (11:05-12:30)

Log analysis, cert roadmap, Python exercise intro, career paths. Parts 3, 4, and 6.

Cyber Range (1:30-2:25)

Kali Linux (2025.7) -- Mini CTF: crypto, web, and forensics challenges.

Interactive Game (2:25-3:20)

Part 5 -- Analyze login logs, identify the breach, write an incident report.

Presentations (3:20-4:20)

5 pairs, approximately 10 min each: present your week project to the group.

Wrap Up (4:20-4:30)

Certificates, celebration, and cybersecurity career resources.